

Advanced Dynare

Chapter XIII

Solving Models with Occasionally Binding Constraints

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Outline

1. Introduction

2. A prototype model

3. Perfect foresight simulations

4. Stochastic simulations: OccBin

Overview

- Focus on models with [occasionally binding constraints](#)
- Since Dynare 5: implementation of OccBin toolkit to solve models with occasionally binding constraints
- Feature only recently added
→ please report any bugs you may encounter

Occasionally binding constraints

- Historically, economists relied on linearized models:
 - Useful to gain intuition
 - Easy to apply, especially for estimation (linear state space tools)
- However: many interesting economic problems and policy questions feature nonlinearities such as **occasionally binding constraints (OBCs)**:
 - effective lower bound on nominal interest rates (Christiano et al. 2011)
 - Credit constraints (Guerrieri and Iacoviello 2017)
 - Downward nominal wage rigidities (Schmitt-Grohé and Uribe 2016)
- Not always sensible to assume OBC is always binding as in e.g. Iacoviello (2005) or Jermann and Quadrini (2012)

Occasionally Binding Constraints in Dynare

- Dynare essentially offers three options to solve models with occasionally binding constraints:
 1. Nonlinear perfect foresight models: brute force (max/min-operators) and mixed complementarity problems (MCP)
 2. Linearized stochastic models with surprise shocks: OccBin
 3. Stochastic nonlinear models: Extended Path

Crucial distinction: informational structure

1. deterministic/perfect foresight: all future shocks are known exactly
2. stochastic simulations: only the shock distributions are known

- Deterministic:

- + Arbitrary precision
- + Easily captures nonlinearities (such as OBCs)
 - Assumption of perfect foresight
 - How to simulate time series?

- Stochastic:

- + Captures uncertainty
- + Needed for estimation with state space form
 - Typically suffers from curse of dimensionality
- + Perturbation methods can capture OBCs with additional tools (OccBin)
 - Often uses 1st order approximations: certainty equivalence

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A basic New Keynesian model

- Consider a three equation linearized New-Keynesian textbook model (e.g. Galí 2015; Walsh 2010)

1. New Keynesian Phillips curve

$$\pi_t = \beta E_t [\pi_{t+1}] + \kappa y_t + \varepsilon_t^a \quad (1)$$

2. New Keynesian IS curve

$$y_t = E_t [y_{t+1}] - \frac{1}{\sigma} (i_t - E_t [\pi_{t+1}]) + \varepsilon_t^c \quad (2)$$

3. An interest-rate rule

$$i_t = \rho_i i_{t-1} + (1 - \rho_i)(\phi_\pi \pi_t + \phi_y y_t) + \varepsilon_t^i \quad (3)$$

- The shock processes are AR(1):

$$\varepsilon_t^j = \rho^j \varepsilon_{t-1}^j + u_t^j, \quad u_t^j \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_j^2) \quad \forall j \in \{a, c, i\} \quad (4)$$

Our OBC: the Effective Lower Bound

- An **effective lower bound** (ELB) on nominal interest rates constrains central banks from actually lowering interest rates below i^{lb}
- Modify our equation to account for the nonlinearity:

$$i_t = \max\{i_t^{not}, i^{lb}\}, \quad (5)$$

where the notional interest rate i_t^{not} in the absence of the ZLB is given by

$$i_t^{not} = \rho_i i_{t-1}^{not} + (1 - \rho_i)(\phi_\pi \pi_t + \phi_y y_t) + \varepsilon_t^i \quad (6)$$

- We will use demand shock u_t^c to induce ELB to bind

Dynare code: Core part of the model

```
model(linear);  
[name='Phillips Curve']  
pie=beta*pie(+1)+kappa*y+zeps_a;  
  
[name='IS curve']  
y=y(+1)-1/sig*(inom-pie(+1))+zeps_c;  
  
[name='Notional rate Taylor rule']  
inomnot=rhoinom*inomnot(-1)  
      +(1-rhoinom)*(phi_pi*pie+phi_y*y)+zeps_i;
```

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Perfect foresight: brute force ELB implementation

- ELB implementation via max-operator in the model-block.

```
[name='Notional rate Taylor rule']  
inomnot=rhoinom*inomnot(-1)  
        +(1-rhoinom)*(phi_pi*pie+phi_y*y)+zeps_i;  
  
[name = 'Observed interest rate']  
inom = max(inomnot,ilb);
```

- Of note: max-operator is not differentiable
→ challenging for Newton-type solvers that rely on (sparse) Jacobian

Simulating a demand shock of 1%

```
shockssequence = -0.01;  
  
shocks;  
    var u_c;  
    periods 1;  
    values(shockssequence);  
end;  
  
perfect_foresight_setup(periods=50);  
perfect_foresight_solver;
```

- Important: allow for sufficiently many periods to return back to steady state

Perfect foresight: mixed complementarity problem (MCP) implementation

- Alternative: use specialized mixed complementarity problem (MCP) solver:
 - Levenberg-Marquardt mixed complementarity problem solver (Kanzow and Petra 2004)
triggered by keyword `lmmcp` or
`perfect_foresight_solver(stack_solve_algo=7, solve_algo=10)`
 - PATH solver (Ferris and Munson 1999):
`perfect_foresight_solver(stack_solve_algo=7, solve_algo=11)`
- Solvers require particular type of setup:

$$LB = X \quad \Rightarrow F(X) > 0, \quad (7)$$

$$LB \leq X \leq UB \quad \Rightarrow F(X) = 0, \quad (8)$$

$$X = UB \quad \Rightarrow F(X) < 0. \quad (9)$$

- Sign of the equation residual with binding constraint matters
- Convention in Dynare: LHS-RHS

Perfect foresight: MCP implementation

- Explicitly specify a complementarity condition for the Taylor rule using the perpendicular symbol:
 - $\perp$ UTF-8 symbol (Unicode codepoint U+27C2)
 - `_|_`: pure ASCII, i.e. a vertical bar enclosed within two underscores

```
[name='Notional rate Taylor rule']  
inom=inomnot  $\perp$  inom>-0.0055;
```

```
[name='Notional rate Taylor rule']  
inomnot=rhoinom*inomnot(-1)  
+(1-rhoinom)*(phi_pi*pie+phi_y*y)+zeps_i;
```

- If lower bound on interest rate is binding

$$residual = i_t - i_t^{not} > 0 \quad (10)$$

Perfect foresight: MCP implementation

- Alternative: older syntax where mcp-tag indicates equation to be replaced (RHS expression is not parsed!)

```
[name='Notional rate Taylor rule',mcp='inom>-0.0055']  
inom=inomnot;
```

```
[name='Notional rate Taylor rule']  
inomnot=rhoinom*inomnot(-1)  
+(1-rhoinom)*(phi_pi*pie+phi_y*y)+zeps_i;
```

- If lower bound on interest rate is binding

$$residual = i_t - i_t^{not} > 0 \quad (11)$$

Simulating a demand shock of 1%

```
shockssequence = -0.01;  
  
shocks;  
    var u_c;  
    periods 1;  
    values(shockssequence);  
end;  
  
perfect_foresight_setup(periods=50);  
perfect_foresight_solver(lmmcp);
```

- Specialized solver triggered by keyword `lmmcp`

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Overview

- Dynare applies a [perturbation](#) approach to solve stochastic model: Taylor expansion
- But: OBCs introduce non-differentiability
- OccBin (Guerrieri and Iacoviello 2015): handle OBCs using a piecewise linear approach
- OBCs are different regimes of the same model, e.g.:
 - economy outside the ELB
 - economy at the ELB
- The solution can be highly nonlinear due to feedback loop:
 - dynamics in one regime depend on its expected duration
 - in turn, the expected duration depends on the states

Strengths and weaknesses

- Similar to the linear solutions, the method does not take into account uncertainty (and (precautionary) behaviour towards it)
→ solution does not take into account that the constraint(s) could be binding in the future (e.g. because of unrealised shocks).
- But:
 - it is quite fast and can be used for estimation
 - can be applied to very large models

Representing two regimes

- The reference/baseline regime (e.g. ELB constraint is slack) can be expressed as a linearized system:

$$A_1 E_t [X_{t+1}] + A_0 X_t + A_{-1} X_{t-1} + \mathcal{E} u_t = 0, \quad (\text{M1})$$

where X_t denotes the model variables in deviation from steady state.

- The alternative regime (e.g. ELB constraint is binding) features a constant D^*

$$A_1^* E_t [X_{t+1}] + A_0^* X_t + A_{-1}^* X_{t-1} + D^* + \mathcal{E}^* u_t \quad (\text{M2})$$

- Both models [M1](#) and [M2](#) are linearized around the same point (D^* accounts for potentially different steady states)
- Important: reference regime is the one where the system settles without any shocks

Regimes and solutions

Reference regime (M1)

- Solution can be written as:

$$X_t = \mathcal{P}X_{t-1} + Cu_t \quad (\text{M1DR})$$

- Transition matrices $\mathcal{P}$ and C are constant
- Blanchard-Kahn conditions hold
- Provides linearization point

- Assumes certainty equivalence: system does not depend on u_{t+j} for $j \geq 1$
→ people expect future shocks to equal their expected value of 0 with certainty

Alternative regime (M2):

- Solution can be written as:

$$X_t = \begin{cases} \mathcal{P}_t X_{t-1} + R_t + C_t u_t & \text{for } t = 1 \\ \mathcal{P}_t X_{t-1} + R_t & \text{for } t \geq 2 \end{cases} \quad (\text{M2DR})$$

- Time-varying matrices depending on expected regime length

Solution strategy: Backward induction + Guess and verify

- We know terminal condition: if $u_t = 0$ for all future periods, system by construction returns to reference regime M1
→ need to check whether we are indeed back in M1 from T onwards (i.e. the constraint is slack)
- For $t < T$: Regime M2 given state X_{t-1} (and potentially shocks u_t)
- Guess and verify approach to find regime length:
 1. Guess T in period t_0 such that for $t \geq T$, we are back in M1
 2. Check whether guess is consistent with implied model dynamics
 3. If not, update guess

Working backwards: the terminal period T

- For any $t \geq T$ with expected $u_t = 0$, the solution in regime $M1$, ([M1DR](#)), implies:

$$X_t = \mathcal{P}X_{t-1} \quad (12)$$

- For expectations in $T - 1$ then follows:

$$E_{T-1} [X_T] = \mathcal{P}X_{T-1} \quad (13)$$

- This allows eliminating the forward-looking part of the linear system for $M2$:

$$\begin{aligned} A_1^* E_{T-1} [X_T] + A_0^* X_{T-1} + A_{-1}^* X_{T-2} + D^* &= \\ A_1^* \mathcal{P}X_{T-1} + A_0^* X_{T-1} + A_{-1}^* X_{T-2} + D^* &= 0 \end{aligned} \quad (14)$$

- Given inherited states X_{T-2} , this is a matrix equation in one unknown: X_{T-1}

Working backwards: the solution for $T - 1$

- Solve for X_{T-1} given X_{T-2} to get the decision rule:

$$X_{T-1} = -(A_1^* \mathcal{P} + A_0^*)^{-1} (A_{-1}^* X_{T-2} + D^*) \quad (15)$$

- Looks complicated, but is actually linear:

$$X_{T-1} = \underbrace{\mathcal{P}_{T-1}}_{-(A_1^* \mathcal{P} + A_0^*)^{-1} A_{-1}^*} X_{T-2} + \underbrace{R_{T-1}}_{-(A_1^* \mathcal{P} + A_0^*)^{-1} D^*} \quad (16)$$

- Linearity allows for easy evaluation of conditional expectations $E_{T-2}[X_{T-1}]$ required in $T - 2$

Working backwards until the first period

- Continue the drill with the previous period $T - 2$:

$$A_1^* \mathcal{P}_{T-1} X_{T-2} + A_0^* X_{T-2} + A_{-1}^* X_{T-3} + D^* = 0 \quad (17)$$

- Solve for X_{T-2} given X_{T-3} to get the decision rule:

$$X_{T-2} = \mathcal{P}_{T-2} X_{T-3} + R_{T-2} \quad (18)$$

with

$$\mathcal{P}_{T-2} = - (A_1^* \mathcal{P}_{T-1} + A_0^*)^{-1} A_{-1}^* \quad (19)$$

$$R_{T-2} = - (A_1^* \mathcal{P}_{T-1} + A_0^*)^{-1} D^* \quad (20)$$

- $\mathcal{P}_t, R_t$ and later C_t are time-varying matrices composed of structural matrices in M1 and M2

→ nonlinear dynamics

Working backwards: the first period

- We finally reach the first period $t = 1$:

$$A_1^* \mathcal{P}_2 X_1 + A_0^* X_1 + A_{-1}^* X_0 + D^* + \mathcal{E}^* u_1 = 0 \quad (21)$$

- Solve for X_1 given X_0 and current shock u_1 to get the decision rule:

$$X_1 = \mathcal{P}_1 X_0 + R_1 + C_1 u_1 \quad (22)$$

with

$$\mathcal{P}_1 = -(A_1^* \mathcal{P}_2 + A_0^*)^{-1} A_0^* \quad (23)$$

$$R_1 = -(A_1^* \mathcal{P}_2 + A_0^*)^{-1} D^* \quad (24)$$

$$C_1 = -(A_1^* \mathcal{P}_2 + A_0^*)^{-1} \mathcal{E}^* \quad (25)$$

- This completes the computation of the solution ([M2DR](#))

Verification

- We now have obtained time-dependent decision rules for all periods, but conditional on potentially wrongly guessed T
- Is behavior of economy starting from X_0 with given shock ε_1 consistent with conjectured decision rules?
→ compute (time) path for X_{t+j} and implied reversal to the baseline regime $\tilde{T}$
- If $\tilde{T} = T$ is verified: we have found a solution!
- If not, update the guess for the regime sequence and back to square one
 - Make use of the computed path for X_t to update the next guess (default)
 - Other updating schemes are possible (e.g. step-wise changes of the regime guess: `simul_curb_retrench`)
- Multiple solutions: multiple regime sequences possible (Holden 2022)

OccBin model block

- Equation name tags are used to name equations
- `bind` and `relax`-tags assign equation pairs to the alternative and the reference regime, respectively, associated with the constraint named `zlb`

```
[name='Notional rate Taylor rule']
inomnot=(1-rhoinom)*inomnot*(-1)
        +rhoinom*(phi_pi*pie+phi_y*y)+zeps_i;

[name = 'Observed interest rate',relax='zlb']
inom = inomnot;

[name = 'Observed interest rate',bind='zlb']
inom = ilb;
```

OccBin constraints definition

- Declare regime switching condition

```
occbin_constraints;  
name 'zlb'; bind inomnot<=ilb;  
end;
```

which is equivalent to

```
occbin_constraints;  
name 'zlb'; bind inomnot<=ilb; relax inomnot>ilb;  
end;
```

- Explicit relax-tags are required when bind-expression cannot be evaluated in alternative regime

OccBin constraints: numerical considerations

- In each simulation period, Dynare will check the respective conditions to see whether a regime change takes place
→ numerical imprecisions can cause spurious changes at the boundary between regimes
- Useful trick: introducing a safety buffer:

```
occbin_constraints;  
name 'zlb'; bind inomnot<=ilb-1e-8; relax inomnot>ilb+1e-8;  
end;
```


Defining surprise shocks

```
shockssequence = -0.01;  
  
shocks(surprise);  
    var u_c;  
    periods 1;  
    values(shockssequence);  
end;
```

- Keyword surprise of shocks-block indicates that perfect foresight syntax is used, but future shocks are not anticipated

Triggering OccBin simulations

```
occbin_setup; //instruct OccBin use  
occbin_solver(simul_periods=20,simul_check_ahead_periods=50);  
occbin_graph y inom inomnot pie;
```

- `occbin_setup` is mandatory to trigger the use of the OccBin toolkit (translates `shocks(surprise)-block`)
- `occbin_solver` triggers actual simulation run
 - `simul_periods`: number of periods for which to compute simulation
 - `simul_check_ahead_periods`: initial guess for period T at which regime returns to baseline

OccBin output

- Dynare stores all OccBin simulation-related output in `oo_.occbin.simul`
 - the simulated solution paths in `oo_.occbin.simul.endo_piecewise`
 - the linear simulation when ignoring the constraints in `oo_.occbin.simul.endo_linear`
 - an error-flag in `oo_.occbin.simul.error_flag`
 - the regime and its expected duration in `oo_.occbin.simul.regime_history`
- Details can be found in the manual

OccBin: nonlinear models

- Dynare's implementation can handle non-linearly entered models
→ Dynare will do the linearization
- Attention: expressions within `occbin_constraints`-block are not linearized (yet)
→ risk for inconsistency

Literature I

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Literature II

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- Walsh, Carl E. (2010). *Monetary theory and policy*. MIT Press.